



Hanford

Introduction

The AI industry is approaching a ceiling. It's a norm that the community has settled into that, I personally believe, will hold further progress back into another AI winter, unless a company goes back to the drawing board and offers a brand new service given the observed use cases of the technology.

These norms can be categorized into two overarching issues. First, **the affinity of the industry to transformer-based architectures is antithetical to the AI Agent use case**. Agents have been poised to replace long research tasks across consumer, healthcare, finance, academia, and many other verticals. These industries require agents to run for undetermined lengths, going as deep as possible into a subject or task and reporting back, in detail, on its findings. Transformer based architectures have the attention matrix to thank for its preference in the space, but that has come at a disadvantage via context window lengths. Most frontier models today are extremely compute-inefficient, requiring a full re-computation of the attention matrix for each new token generated ($O(s^2)$ memory complexity), and are capped at a context limit of 128-256k tokens, amounting to 400-500 pages[1], not enough for an average research payload of 1000-2000 pages.

Second, **AI Agents have been designed to take from your attention plate, not your workload plate**. Most AI agents offered by frontier AI companies present you with a chat interface and the ability to integrate different data sources into it for more customized responses. Ultimately, it amounts to yet another tab in your browser. Coding aside, AI agents have yet to leave their chat interfaces and work where you do at scale. In other words, there is no agent in the space capable of meeting you where you already work; in your email inbox, on slack, in your texts, and is capable of stitching the data from each together for one amalgamated context of your workday.

Overview

Hanford will address the two issues mentioned above through two intertwined efforts; via the Hanford Agent Platform, that will create an agent optimized for working outside of a chat window, as well as Hanford Research, which will get to work building the first Agentic model, one with ultra-long context windows for long-form use cases, and equivalent reasoning and tool use capabilities to match today's leading language models.

Hanford Agent Platform

Diving into further detail of the Hanford Agent Platform, our efforts will be focused on addressing the following issues that exist in the industry. In other words, the solutions to the following points will be Hanford's product roadmap:

1. **The “Friend” Fallacy** – AI agent products like ChatGPT have been fine-tuned to treat you like a friend rather than someone asking for new perspectives or for their thoughts to be challenged. Specifically in the consumer space, frontier companies have realized that the new definition of a sticky product in this age is the measure of an agent’s sycophancy towards its users, not the effectiveness of the agent in completing tasks. ***Hanford’s Agent will be a professional assistant that does not provide opinions, but acts as an extension of your professional life.***
2. **Unnatural UIs** – For products where the main point of interaction is a conversation, all of these products have introduced very unnatural UIs involving multiple chats, and freemium limits applied at the most anger-inducing points of the usage process. This can partly be blamed on the now-accepted model architecture, which limits the lengths of all conversations. ***Hanford’s Agent will be designed to be communicated with via a centralized platform, email, and text, much like a human assistant.***
3. **Caged Potential** – After building models capable of reasoning, tool calling, as well as deep research, these products have caged the capabilities of these models to their apps only. To highlight an already addressed example, cursor has shown the benefit of allowing an agent to exist alongside your code, building and improving what one has done. Unless the masses create new companies and VCs pour millions into them to figure out the rest of the tools that we use on a daily basis (emails, excel files, databases, etc.), there will be a never-ending quest to build a true assistant, and that will involve tens of different subscriptions that will create more overhead than the pain it is trying to solve. ***Hanford’s Agents will contain the ability to self-extend, building their own tools for your specific use case to meet you where you are working, regardless of where you are.***
4. **Retention Failures** – Through a combination of the issues already mentioned above, these models create a paradox where, while agreeing with the user on every note and acting as their friend, it fails to remember who they are. This leads to an awkward experience equivalent to opening a new chat with someone you’ve been talking to for a few weeks, and needing to re-introduce yourself to them. These products fail to gain deep understandings of not only the user, but the user’s life; things such as response tone/language, open loops, and emotions all go unprocessed. ***Hanford’s Agent will contain a collection of background processes that review, learn, and adjust their methods of working with you by studying each day’s message history.***
5. **True Agency** – Finally, the most overarching theme of them all. No matter how intelligent these agents and models are, they do not speak unless they are spoken to. This is possibly the grandest missed opportunity of a product claiming to be an assistant or extension of the user; it simply cannot act on its own given new information, and instead waits for the user to provide the information for it to respond. A true product that intends to take away from the plate of a user must act as an extension of the user’s capability in **receiving information**. It must be able to, for example, field an email on

behalf of the user, schedule a meeting, reschedule a meeting, handle access of a data room to a user trying to fundraise, etc. The industry has been busy building friends that aim to take away from the wrong plate: the plate of attention. ***Apart from user input, Hanford will build itself a collection of listeners; webhooks that inject new data into its context in order for it to alert you and recommend actions.***

Hanford Research

While it won't be a blocker for this effort, the usage data from the early versions of the agent platform are going to feed a separate team within the company tasked with researching and building Hanford Mk. 1, an SSM-based language model with super-extended context windows matching the reasoning and tool calling capabilities of today's frontier models.

Much work has already been done in building State Space Models such as Mamba and S4, but core issues of these models need to be visited by community researchers due to the lack of a VC-backed effort in establishing and serving these architectures for feedback and adjustment. The research work will be driven by the following pillars:

1. **Deviation of Creative Destruction** – the demand of these compute-inefficient models has fueled the rapid rise of inference centers packed with GPUs that have put an immense amount of strain on our infrastructure. Infrastructure has since become a big factor in determining the winner of the AI race, which has led countries like the US to build them in places that have poisoned water, air, and other basic life factors that any society needs to function.
2. **Solutionism** – When these models were first created, it was never known what the primary use case would be. Over time, it became clear that agentic products made up the majority of the use cases for LLMs. The issue in the process was, LLMs and their architectures were solidified before the use case was figured out. Since then, as larger institutions developed MoE and other iterations of transformer-based architectures, only a handful of communities have looked beyond this architecture, solving fundamental problems in novel architectures to enable true agentic use cases. One can look at state space models (SSMs) as a possible option, with Manus founder Yichao Ji saying that “Unlike Transformers, SSMs lack full attention and struggle with long-range backward dependencies. But if they could master file-based memory—externalizing long-term state instead of holding it in context—then their speed and efficiency might unlock a new class of agents. Agentic SSMs could be the real successors to [Neural Turing Machines](#).”

Business Opportunity

Market Size & Growth

- Autonomous AI & Agents have been valued at ~\$8billion in 2023, and are projected to see a >10x increase to \$86.9 billion by 2032 ([link](#))
- Narrowing it down to the personal AI assistant market, specifically in productivity tools for professionals, is projected to grow to \$56.3 billion by 2034 ([link](#))
- The numbers above suggest large growth potential, but also a fast-growing and increasingly competitive space in some respects of the system.

Market Segments & Dynamics

- Market for management productivity use cases (tool orchestration, comms management, workflow automation) are the fastest growing, and show a lot of potential upsell.
- Transition of demand from single-prompt agents to workflow automation continues to grow in the enterprise space.

GTM Plan

- Focusing initial go-to-market on agentic platform power-users, with a focus on roles within organizations with complex workflows (product/engineering managers, venture capital investors)
- Emphasizing the “forever” nature of the platform: memory, continuous monitoring, and recursive self-tool integration.
- Basic freemium subscription model, with access to a basic forever agent for free and a paid version including the tool generation and connectors.
 - A note on the economics – as the research on the model is being done, we’ll most likely need to rely on transformer based models with in-house memory management functionality. Focus on positive cash flow is going to be once hardware requirements are figured out for the model to serve as the agent’s backend.
- At a \$40/mo subscription for unlimited use (undercutting competitors), Hanford will see revenues grow from \$4.8M ARR in year 1 to upwards of \$500M by year 5.

Possible Expansion Plans

- Enterprise – bulk deals for organizations with large head counts, forward deployed support engineers will assist with onboardings + requests and issues.
- Infrastructure – provide Hanford agent as an API for emerging AI browser market, or other use cases in other products.
- Academia – assisting with longer form research and discovery projects

- Policy – providing Hanford to staffers to conduct long-range research on policy issues (treading carefully, of course)

Product Roadmap (For First Funding Round)

Prototyping (ACTIVE - Q4 '25)

- In the prototyping phase, core aspects of the agent are going to be tested out in isolated experiments. This includes:
 - Infinite chat – the SSM to truly enable this is going to take some time (and lots of funding/revenue) to research, so I will start with the building of a memory management system that could sit atop a normal, transformer based model.
 - Recursive tooling – testing out the ability for the agent to generate and then use its own tools.
 - System prompt updates – allowing the agent to study messages from the past 24h, and update its tone or approach to the user
 - “Ambient” processing – listening for emails or other signals and automatically updating the context.
 - Mode switching – switching from email mode to text mode to natural (on the web app) mode
- This phase will also consist of picking the models that are going to be making up Hanford's different capabilities (e.g. sonnet for conversation, codex for tool building, llama for function calling, etc.)

MVP (Q1 '26)

- The MVP phase is going to focus on tying the prototypes together into a single web application, which will consist of:
 - Initial architecture setup
 - Model integrations
 - Design and implementation of UI (which will take a long time)
 - Development of a set of known connectors (email clients, github, etc.)
 - Early product analytics for tracking stickiness.
- A handful of connections considered ICPs will be tapped to try out this alpha version of the application, with weekly meetings held for feedback.
- The main reason for the 3 month timeframe is that it will also consist of hiring the team to build the remainder of this platform.

Invite-Only Beta (~Q1 + Q2 '26)

- Following the processing of all the feedback from the MVP phase, time will be taken to make necessary fixes and adjustments.

- Then, we will begin our first push for controlled expansion of our user base. This will involve the following:
 - Implementation of an invite/referral system (incentives TBD).
 - Public changelog for early users to track updates to Hanford.
 - Public (to users only) roadmap where users can request and vote for features.
- On the product development side, we will also begin to blueprint and implement an early version of the Hanford Desktop Agent, which will be a background item that can run on your device (e.g. laptop) that allows for Hanford to execute actions on your device should it see fit.
 - An example of a use case for the desktop agent can be seen in pillar 6.
- The extra quarter will be a way to temporarily push down focus on infrastructure as the team focuses primarily on functionality of the product itself.

Hanford V1 (Q3 + Q4 '26)

- This phase is going to consist of the following:
 - System + Architecture Optimizations
 - General availability of Hanford, and subsequent freeze of the invite system.
 - Implementation of first version of the pricing plan, high love MVP users and first-order invitees grandfathered as product champions.
 - Release of the Hanford Desktop Agent for added functionality of the platform.
- This quarter will generally consist of “being on our toes” to handle the scale of GA.

Hanford V2 (H1 '27)

- Main focus will be the implementation of Hanford Mk1, first model released from the research team, while continuing to address customer requests and issues that need to be fixed.

Connects

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[1]

		Model	Model																																							
			GPT-2 (137M)	mamba-2-8b-hf	nkv-6-world-7b	v5-Eagle-7B-HF	gemma-2-9b-it	Meta-Llama-3-8B-Instruct	recurrent-gemma-9b-it	LLaMA-2-7B-32K	longchat-7b-v1.5-32k	Llama-2-7B-32K-Instruct	01-ai/Yi-34B	Mistral-7B-Instruct-v0.3	Mistral-8x7B-Instruct-v0.2	Mixtral-8x22B-Instruct-v0.1	activation-beacon-lama2-7b-chat	Yann-Mistral-7b-128K	chatglm3-6b-128K	activation-beacon-mistral-7b	Phi-3-mini-128K-instruct (3.8B)	Phi-3.5-MoE-Instruct (3.8B)	Phi-3.5-MoE-Instruct (3.8B)	Phi-3.5-MoE-Instruct (3.8B)	Meta-Llama-3-1-8B-Instruct (6.6B/6.6B)	gpt-4o-mini-2024-07-18	Owen2-5-TB-Instruct (14B)	GPT-4 (0125-preview)	Meta-Llama-3-1-70B-Instruct	Owen2-5-TB-Instruct	Gemini 1.5 Pro	Llama3-ChatQA-1.5-8B + RAG	RMT (137M) fine-tune	-ARMT (137M) fine-tune								
Input size	0K	27	70	56	62	72	64	62	41	46	48	49	72	59	60	65	65	75	55	51	56	59	52	64	65	70	64	67	72	67	74	68	87	85	83	98	92	48	99	99	OK	
	1K	15	52	55	54	71	60	59	53	42	47	52	52	60	56	63	59	73	52	52	55	56	55	57	53	70	64	68	70	68	72	68	81	81	78	99	88	48	97	96	1K	
	2K		35	48	48	69	58	57	45	40	46	49	43	55	52	60	56	70	47	43	51	51	48	55	50	62	63	66	67	66	71	68	77	78	76	99	82	47	95	98	2K	
	4K		9	35	41	64	50	21	40	41	43	43	37	45	49	55	54	65	43	40	48	48	46	51	48	59	61	66	62	65	66	65	74	74	75	99	78	46	92	98	4K	
	8K		0	7	2	0	44	14	39	42	40	40	38	29	45	50	52	58	36	38	46	43	45	50	46	58	59	62	60	62	62	65	71	70	73	99	75	45	90	98	8K	
	16K						8	32	39	39	36	35	31	26	42	46	50	51	23	30	41	37	36	46	45	53	52	60	57	60	60	63	64	65	71	99	72	45	86	98	16K	
	32K						3	5	4	5	4	29	37	40		48	43	16	16	36	36	37	42	41	43	51	56	53	56	54	60	53	59	68	98	68	44	78	98	32K		
	64K														48	35	8	10	21	27	29	37	40	38	46	49	45	49	45	49	45	55	43	53	66	97	64	42	70	98	64K	
	128K																6	9	13	14	24	7	34	10	38	39	30	39	43	45	36	45	58	93	59	45	59	97	128K			
	512K																																					51	42	46	95	512K
1M																																					42	39	43	93	1M	
10M																																					37	34	77	10M		
avg		21	33	40	41	55	55	37	36	36	38	39	40	43	49	54	54	59	32	32	41	41	41	41	45	47	51	55	59	58	59	61	62	65	68	72	98	70	44	74	95	avg

While today's transformer-based models tout long context windows, model recalling capabilities within these windows differ between each lab. Models like GPT-4 and Alibaba's Qwen-2.5 might say that they have a 128k token context window, but looking at the results of the Needle In A Haystack evaluations of these models, recalling scores drop significantly after reaching the midpoint of their context windows.

What this means is that, even with the tens of companies that have come out in the past few months claiming to give these models external file memory systems, where in the context window these recollections are provided matter. Additionally, the aforementioned realistic estimate of 400-500 pages earlier in this paper is reduced by %25-%50.

On the other hand, two models on this chart stick are outliers, ones that have oddly similar architectures:

1. Mamba (130M) fine-tune: a pure SSM based model using HiPPO state matrices rather than attention matrices to recall previous steps.
2. ARMT (137M) fine-tune: a hybrid SSM with an attention matrix that was able to extend its context to 10M tokens with an average of %95. Compared to GPT-4's %65 over a 128k token payload, one could see how certain components of SSMs are the true unlocking trait of model context windows.

[2] Competitive Analysis

On agent functionality, my competitors are Runnable and Manus. These two companies fall in line with the design principles set by ChatGPT that is the agent solely existing on the platform, taking away from a user's attention and just making a user's workload more tolerable, not reducing it.

On agent tone and service, my competitor is Poke. While Poke also lives where you work, it serves as a personal companion rather than a professional assistant, which many users have reported on due to its always-sarcastic tone. In terms of functionality, Poke only covers a limited set of personal tools like delivery apps, maps, and emails, and does not have the ability to build its own tools.

My most direct competitor to date is dimension.dev, a new competitor that has seemed to understand this market. In the short term, they lack much of what Hanford already has, like self-extension and comms optimizations. In the long term, their dependence on LLMs will act as a constraint for ever-growing tasks.